

# CHENGZHI YAN

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## RESEARCH INTERESTS

Artificial intelligence and machine-learning applications, with an emphasis on healthcare-related computing. Interested in data processing & engineering, time-series modelling, multi-source data integration, software system construction and AI-driven clinical auxiliary solutions.

## EDUCATION

**M.Sc. in Computer Science, University of Bristol, UK**

**Sep 2026 – Jun 2027 (Expected)**

Relevant Courses: Deep Learning, Sustainable Computing, Advanced Topics in Computer Science, Individual Research Project

**B.Sc. in Computer Science, University of Bristol, UK**

**Sep 2022 – Jun 2026**

Relevant Courses: Artificial Intelligence, Software Engineering, Algorithms, Computer Systems, Computer Architecture

## RESEARCH EXPERIENCE

**AI R&D Intern — Report Quality Assessment System**

**Summer 2024**

*Radtech Imaging Technology (Shenzhen) Co., Ltd.*

- Developed AI-based text recognition and semantic analysis for the Report Quality Assessment System: designed a text rule engine and key-content extraction modules to parse radiology report structure, including body parts, findings, size, count and diagnostic conclusions.
- Built a 170k-entry medical lexicon with jieba word segmentation, Word2Vec pre-trained vectors and extensive regex rules; extracted subjects and compared findings against a lesion dictionary for structural checks.
- Handled negation and uncertainty semantics (e.g., distinguishing "no abnormality seen" from "not mentioned"); upgraded to a second-version semantic recognition pipeline that extracts subject–verb–object triples, with special handling for non-sequential, abbreviated report writing.
- Implemented typo detection based on MacBERT-base with homophone and similar-character recognition ( $\approx 95\%$  accuracy on detected typos), fine-tuning the medical lexicon to reduce false positives.
- Covered 15 quality-control dimensions: missing body parts, terminology misuse (CT-density vs MRI-signal), punctuation errors (Chinese/English mixing), laterality mismatch, examination-method errors (enhanced vs plain scan), contradictory language, measurement errors, gender errors, missing RADS grading, description–conclusion mismatch, report timeliness, typos and sensitive words.
- Evaluated diagnosis–imaging consistency with a large language model and generated medical-history summaries via lesion/organ extraction; gained hands-on experience in medical NLP, rule-engine design and clinical quality-control workflows.

**AI R&D Intern — Image Quality Assessment System**

**Summer 2025**

*Radtech Imaging Technology (Shenzhen) Co., Ltd.*

- Developed AI-based image recognition for the Image Quality Assessment System: body-position recognition, anatomical structure segmentation, radiographic coverage measurement and foreign-body/artifact detection.
- Implemented semantic segmentation with DeepLab v3+ to produce masks, and used OpenCV to compute bounding boxes, positions and angles for structure localisation and measurement.
- Applied data augmentation (left-right flipping, Gaussian blur, HSV colour-space transforms, random brightness/contrast, elastic deformation) to improve recognition robustness; adjusted resolution for body parts barely visible to the naked eye.
- Tackled overfitting caused by cross-hospital data variance (e.g., steel implants and fractures) by explicitly including foreign bodies such as implants in model training; used a 3:1 training–validation split.
- Implemented quantitative scoring of image metrics including information entropy, detail fidelity and noise, building automated image-quality evaluation pipelines.

- Participated in a national medical-insurance chest X-ray multi-disease recognition competition, with a technical approach covering MAE self-supervised pretraining, a ViT backbone, multi-granularity classification heads, disease GCN graph inference and ASL asymmetric loss.

### **Applied Data Science | Early Prediction of Sepsis**

**Sep 2025 – Jun 2026**

*University of Bristol — Supervisor: Prof. Saptarshi Sinha*

- Developed machine-learning solutions for clinical predictive tasks using real-world patient datasets.
- Performed full-cycle dataset processing: data cleaning, missing-value imputation, feature importance analysis and multi-dimensional data quality assessment.
- Built reusable automated data processing pipelines; implemented hybrid time-series models including GRU-D + LightGBM and BRITS + LSTM, outperforming baseline clinical scores on held-out test sets.
- Gained hands-on experience in complex dataset engineering and applying deep-learning techniques to domain-specific challenges.

GitHub: <https://github.com/billu615/Applied-Data-Science-CW-Sepsis.git>

### **Software Engineering | Children's Mental Health Website**

**Sep 2023 – Sep 2024**

*University of Bristol — Supervisor: Prof. Sarah*

- Collaborated to build a web-based resource platform for public-health-related content delivery.
- Designed MongoDB data schemas and unified data entry standards; managed data models and multi-type content assets.
- Implemented front-end modules with EJS and JavaScript; constructed backend services based on Node.js and Express.js.
- Practised end-to-end web development, business-oriented data modelling and structured content management.

GitHub: <https://github.com/spe-uob/2024-ThinkingSpace>

### **Team Project | Unity VR Accessibility Simulation**

**Sep 2025 – Jun 2026**

*University of Bristol — Supervisor: Prof. Sarah*

- Developed an immersive VR simulation targeting accessibility-related research scenarios with Unity.
- Managed heterogeneous assets including 3D scenes, character resources and physiological sensor signals.
- Deployed LSTM-based time-series model for signal detection, integrated hardware tactile feedback via haptic vest.
- Built practical engineering capabilities for multi-source heterogeneous-data integration and interactive system development.

GitLab: [https://gitlab.com/justice-246/btff/-/tree/dev/game?ref\\_type=heads](https://gitlab.com/justice-246/btff/-/tree/dev/game?ref_type=heads)

### **Computer Systems | Conway's Game of Life Implemented in Go**

**Sep 2024 – Jan 2025**

*University of Bristol — Supervisor: Prof. Tilo*

- Took part in project planning, literature review and formal technical report writing.
- Implemented cellular-automaton simulation in Go, including unit testing and system-level performance optimisation.
- Strengthened Go programming, performance tuning, literature survey and academic technical-writing competencies.

GitHub: [https://github.com/chw002/CSA\\_001.git](https://github.com/chw002/CSA_001.git)

## **PATENTS**

### **Granted Utility Model Patents**

#### **1. Medical Radiation Protection Screen — Patent No. ZL202421476455.6 | Granted 2024**

Inventors: C. Yan (1st), X. Huang (2nd) | Contribution: Led mechanical-structure design for medical radiological shielding hardware.

#### **2. Automatic Radiation Protection Screen — Patent No. ZL202120693771.9 | Granted 2021**

Inventors: Y. Yang (1st), C. Yan (2nd), J. He (3rd) | Contribution: Contributed to mechanical-structure and automatic-control design for radiology-protection equipment.

**3. Electromagnetic Relay-Based Switching Control Circuit and Medical X-ray Protection Device — Patent No. ZL202420775618.4 | Granted 2024**

Inventors: D. Qin (1st), X. Huang (2nd), C. Yan (3rd) | Contribution: Participated in control-circuit logic design for medical X-ray protective devices.

**INTERNSHIP**

**Radtech Imaging Technology (Shenzhen) Co., Ltd. — AI Intern** **2024 – 2025**

- Developed AI-based image recognition for the Image Quality Assessment System, including body-position recognition, anatomical structure segmentation, parameter measurement and automated image scoring.
- Built AI-based text recognition and semantic analysis for the Report Quality Assessment System, including a text rule engine, semantic extraction modules and multi-dimensional report rating.

**Radtech Imaging Technology (Shenzhen) Co., Ltd. — Embedded Intern** **2021 – 2024**

- Led the mechanical-structure design of medical radiological shielding hardware as the first inventor — Medical Radiation Protection Screen (Patent No. ZL202421476455.6, granted 2024).
- Contributed to mechanical-structure and automatic-control design for radiology-protection equipment — Automatic Radiation Protection Screen (Patent No. ZL202120693771.9, granted 2021).
- Participated in control-circuit logic design for medical X-ray protective devices based on electromagnetic relays (Patent No. ZL202420775618.4, granted 2024).

**Radtech Imaging Technology (Shenzhen) Co., Ltd. — Full-stack Development Intern** **2021 – 2023**

*Digital Enterprise Inventory Management System (DingTalk Open-Platform Ecosystem)*

- Conducted end-to-end requirement analysis, database schema design and full-stack application development.
- Implemented procurement, warehousing and inventory-control modules leveraging DingTalk API with Node.js, MongoDB and JavaScript.
- Replaced offline manual ledger workflows; boosted warehouse inbound-outbound efficiency by 40% and reduced manual-input errors, enabling unified visualised business-data management.
- Accumulated practical experience building enterprise-grade data-centric systems and database optimisation for medical-related informatics research.

**SKILLS**

**Programming Languages**

Proficient: Python, Go, C#, JavaScript | Skilled: C, Java | Familiar: EJS

**AI & Machine Learning**

Proficient: TensorFlow | Skilled: Scikit-learn, LightGBM | Familiar: Time-series models (GRU, LSTM, BRITS), dataset pre-processing, feature engineering, data-pipeline construction, DeepLab v3+, MacBERT

**Database, Systems & Development Tools**

MongoDB, Express.js, Git/GitLab, Linux, Unity VR Development, Haptic hardware debugging | Data Visualization: Matplotlib, Seaborn

**Research & Soft Skills**

English-language literature retrieval, academic writing, experimental design, comparative experiment, dataset processing, group project coordination

**Language Proficiency**

Mandarin: Native | English: Able to read academic papers and draft independent research reports